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AI-Assisted coding Assignment(9.5)

TASK-1:

The screenshot shows the Visual Studio Code interface with the 'AI assisted coding' extension. The Explorer pane on the left shows a project named 'AI ASSISTED CODING' with four tasks. The main editor displays a Python script for a discount calculator. The Chat pane on the right shows a conversation with the AI assistant, where it has been asked to generate docstring for the `calculate_discount` function. The AI has provided a Google-style docstring with arguments and a return value. The terminal at the bottom shows the execution of the script, which prompts the user for an original price and a discount rate, and then prints the final price after the discount.

```
1 def calculate_discount(price, discount_rate):
2
3     # Multiply the price by the discount rate and divide by 100 to get the di
4     # Subtract the discount amount from the original price to get the final p
5     return price - (price * discount_rate / 100)
6
7 if __name__ == "__main__":
8     # Example usage
9     original_price = float(input("Enter the original price: "))
10    discount = float(input("Enter the discount rate (%): "))
11    final_price = calculate_discount(original_price, discount)
12    print(f"Price after (discount)% discount: {final_price:.2f}")
```

gave a Python function without comments

```
def calculate_discount(price, discount_rate):
    return price - (price * discount_rate / 100).generate line-by-line
comments for the function. Modify the function so that it
includes a docstring in Google-style or NumPy-style format.
Compare the generated comments with your manually written
version
```

Used 1 reference

Here's the function with line-by-line comments and a Google-style docstring:

```
def calculate_discount(price, discount_rate):
    """
    Calculate the discounted price.

    Args:
        price (float): The original price of the item
        discount_rate (float): The discount rate as a percentage

    Returns:
        float: The price after applying the discount
    """
    # Multiply the price by the discount rate and divide by 100 to get the discount amount
    # Subtract the discount amount from the original price to get the final price
    return price - (price * discount_rate / 100)
```

PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\python313\python.exe" "C:\Users\DELL\AI assisted coding\task-1"

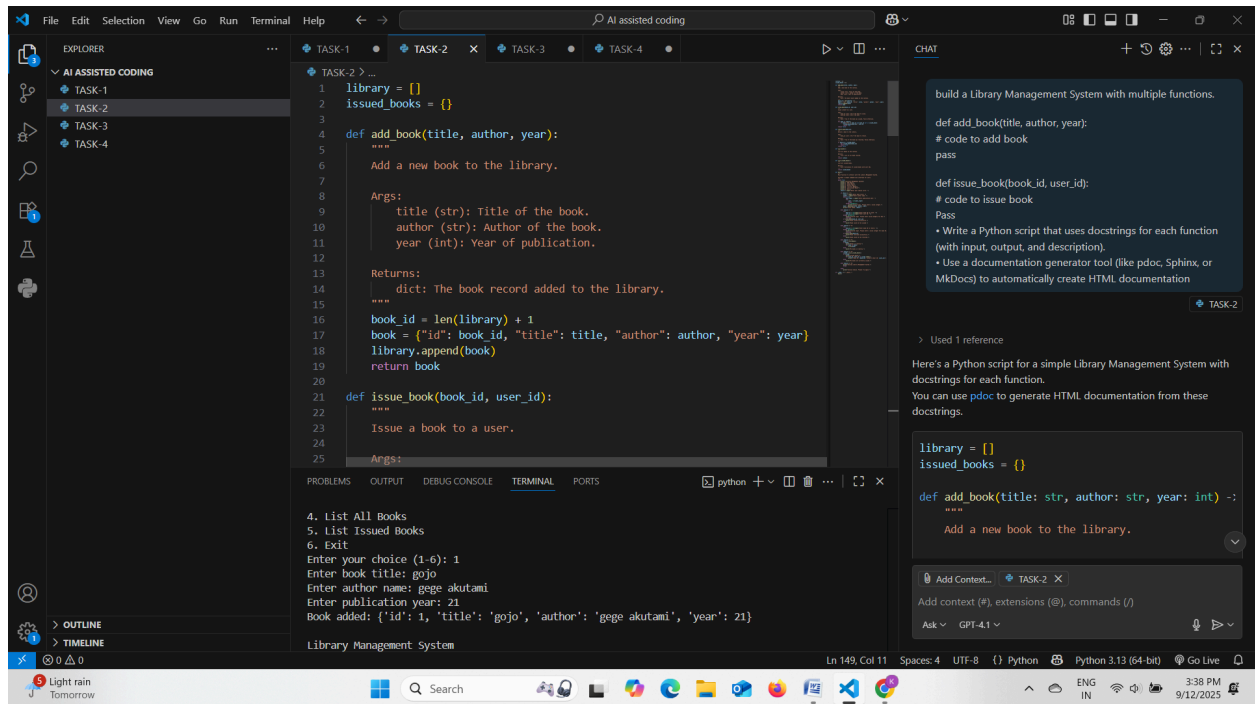
Enter the original price: 159

Enter the discount rate (%): 6

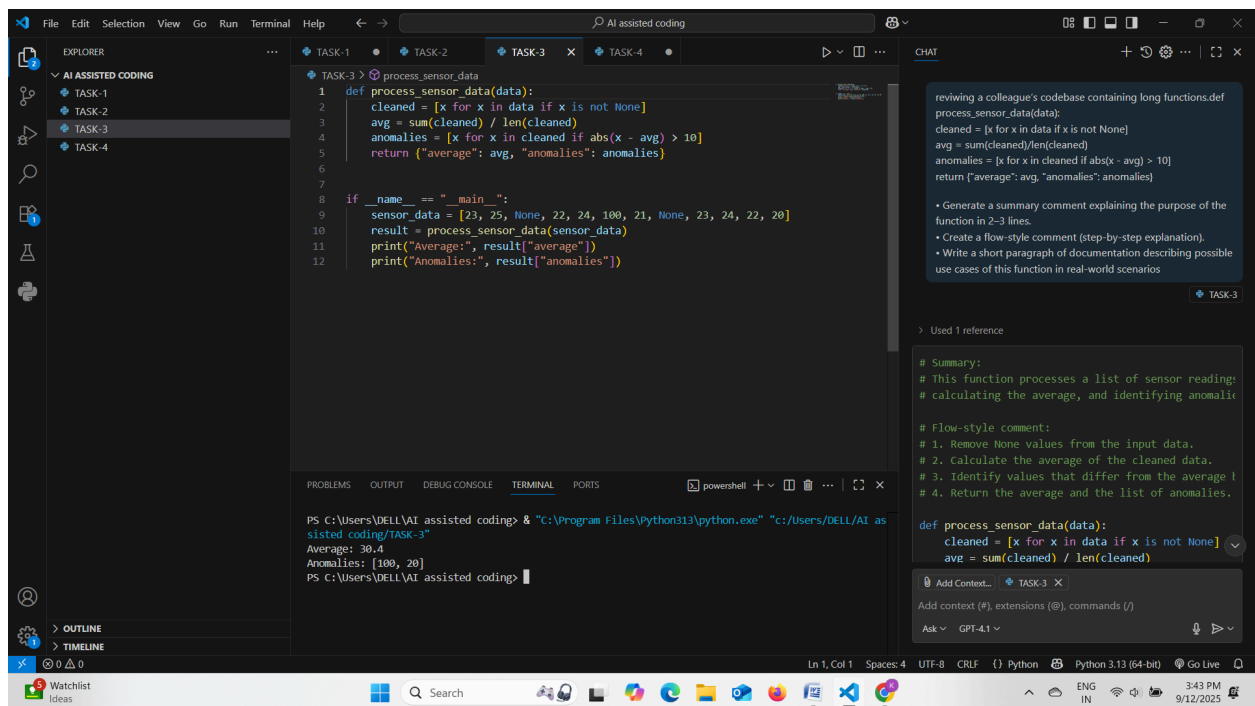
Price after 6.0% discount: 149.46

PS C:\Users\DELL\AI assisted coding>

TASK-2:



TASK-3:



TASK-4:

The screenshot displays the Visual Studio Code editor with a Python chatbot application for TASK-4. The interface is divided into several panels:

- EXPLORER:** Shows the project structure with TASK-1, TASK-2, TASK-3, and TASK-4. TASK-4 is selected.
- Source Control:** Shows the git status for TASK-4.
- Run and Debug:** Shows the current configuration for running the chatbot.
- Code Panel:** Displays the Python code for the chatbot. The code includes a `get_bot_response` function that handles various user inputs like greetings, name queries, and exit commands.
- Chat Panel:** Shows the chatbot's response to the user input. The response is "I'm sorry, I didn't understand that.".
- Usage Guide (from code comments):** Provides instructions on how to use the chatbot, including starting the script, typing messages, and using keywords.
- Terminal:** Shows the output of the chatbot's execution. It displays the chatbot's response to the user input and the user's subsequent interactions.

```
1 # Simple Chatbot Application
2
3 def get_bot_response(user_input):
4     """
5     Returns a response based on user input.
6     Uses basic keyword matching for demonstration.
7     """
8     # Convert input to lowercase for easier matching
9     user_input = user_input.lower()
10
11     # Respond to greetings
12     if "hello" in user_input or "hi" in user_input:
13         return "Hi there! How can I help you today?"
14     # Respond to name queries
15     elif "your name" in user_input:
16         return "I'm your friendly chatbot."
17     # Respond to exit command
18     elif "exit" in user_input or "quit" in user_input:
19         return "Goodbye!"
20     # Default response for unrecognized input
21     else:
22         return "I'm sorry, I didn't understand that. Can you rephrase?"
23
24 if __name__ == "__main__":
25     print("Chatbot is running. Type 'exit' to quit.")
```

Terminal Output:

```
PS C:\Users\DELL\AI assisted coding> & "C:\Program Files\Python313\python.exe" "C:\Users\DELL\AI as
sisted coding\TASK-4"
Chatbot is running. Type 'exit' to quit.
You: hi
Bot: Hi there! How can I help you today?
You: how the weather
Bot: I'm sorry, I didn't understand that. Can you rephrase?
You: can you tell me who is the father of ai
Bot: I'm sorry, I didn't understand that. Can you rephrase?
You: 
```